DGMS 2019 • SECTION E • QUESTION 32

Paleogeography & Global Tectonics: Laurasia, Laurentia & Mesozoic Continental Breakup

Directorate General of Mines Safety (DGMS) Junior Scientific Officer Examination 2019 | Historical Geology & Supercontinent Architecture Guide

Question Statement

Q.32. A major continent that existed before 90 Ma and comprised present-day N. America, Greenland, Europe and part of Asia is referred to as:

a (a) Superia

b (b) Gondwanaland

c (c) Laurentia

d **(d) Laurasia**

Correct Answer: (d) Laurasia (Comprising North America, Greenland, Europe, and Asia)

Official Textbook Figure (Van der Pluijm & Marshak, Earth Structure)

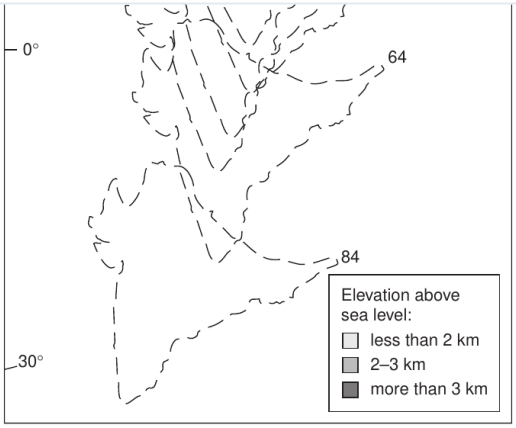


FIGURE 21.2.1 Generalized topographic map of Southeast Asia showing elevation above sealevel (from Burchfiel and Royden, 1991) and positions of India with respect to Asia from Late Cretaceous time until the present. Numbers refer to millions of years before the present. Note the decrease in convergence rate from about 100 mm/y to about 50 mm/y at about 50 Ma.

and Antarctica (called Gondwana). Tethys was perhaps a few hundred kilometers wide in the westernmost Mediterranean region, but widened toward the east, so that, at the longitude of Tibet, Tethys was approximately 6000 km wide.

At the longitude of Tibet, several sizable continental fragments collided with Asia and became accreted

FIGURE 21.2.2 Major fragments in Southeast Asia during Late Paleozoic, Mesozoic, and Early Cenozoic time. The sutures between fragments are shown by barbed lines with the barbs indicating the upper plate of the subduction system. The suture between the Qangtang and Songpan Ganze (SG) fragments is of Mesozoic age and is thought to represent the closure of the Paleotethys Ocean. All sutures north of this are Paleozoic. The suture south of the Lhasa block is the Indus-Tsangpo suture, and represents the closure of Neotethys. A, island arc fragments, V, location of Cretaceous-Early Tertiary arc volcanism. Location of cross section in Figure 21.2.3 is also shown.

to the Asian continental margin before the main collision between India and Asia (Figure 21.2.2). These continental fragments were rifted from Gondwana and moved northward. As each continental fragment was rifted from Gondwana, it opened a new ocean region between itself and Gondwana. At the same time, the old ocean north of each fragment was closed by subduction. For example, the first fragment to collide with Asia closed the early Mesozoic Tethys (more properly called Paleotethys). The Indian subcontinent is the last fragment to have rifted off of Gondwana and as it moved northward, it opened the Indian Ocean to the south of India and closed the Neotethyan ocean to the north. It is the closure of this Neotethyan ocean and the collision of India with Asia that has produced the 2500-km long Alpine-Himalayan mountain chain, along which continental collision and convergence are occurring today, and many parts of it are still incom-

Figure 1: B. A. van der Pluijm & S. Marshak, Chapter 21, Figures 21.2.1 & 21.2.2. Paleogeographic and tectonic map showing the northern composite continent **Laurasia** (consisting of North America, Greenland, Europe, and Asia) separated by the Tethys Ocean from Gondwanaland to the south.

In-Depth Option Evaluation & Paleogeographic Analysis

(d) Laurasia — [CORRECT]

Constituents & Age: Following the initial rifting of Pangea during the Early Mesozoic (~180–150 Ma), the northern continental landmass was termed **Laurasia**. It comprised:

- North America & Greenland (Laurentian craton)
- Europe (Baltica and Avalonian terranes)
- Asia (Siberian, Kazakh, North China, South China cratons, excluding peninsular India)

Laurasia existed as a coherent northern supercontinent until the Late Cretaceous (~90–60 Ma), when the opening of the North Atlantic Ocean and Norwegian Sea finally separated North America/Greenland from Eurasia.

(c) Laurentia — [CRITICAL EXAM DISTRACTOR / INCOMPLETE]

Distinction from Laurasia: Laurentia refers strictly to the ancestral **North American cratonic shield** (North America, Greenland, and parts of NW Scotland). It does **NOT include Europe or Asia**! Laurasia = Laurentia + Eurasia.

(b) Gondwanaland — [INCORRECT • SOUTHERN HEMISPHERE]

Gondwanaland was the southern supercontinent comprising South America, Africa, Madagascar, India, Antarctica, and Australia. It was separated from northern Laurasia by the Tethys Seaway.

(a) Superia — [INCORRECT • ARCHEAN CRATON]

Superia is an ancient Neoarchean supercraton (~2.7–2.5 Ga) formed by the temporary assembly of the Superior, Hearne, Wyoming, and Karelian cratons. It predates the Mesozoic by over 2.3 billion years!

Major Global Supercontinents & Cratonic Amalgamations

Landmass	Constituent Landmasses	Geological Era / Age	Hemisphere / Location
Laurasia	**N. America + Greenland + Europe + Asia (excl. India)**	**Mesozoic (200 Ma to ~90–65 Ma)**	**Northern Hemisphere**
Laurentia	North America + Greenland (+ NW Scotland)	Paleoproterozoic to present (Ancient craton)	North American nucleus only
Gondwanaland	South America + Africa + India + Antarctica + Australia	Neoproterozoic to Mesozoic (~550 to 150 Ma)	Southern Hemisphere
Pangea	Laurasia + Gondwanaland (All global continents assembled)	Late Paleozoic to Early Mesozoic (~335 to 175 Ma)	Global supercontinent
Superia	Superior + Hearne + Wyoming + Karelian cratons	Neoarchean (~2.7 to 2.45 Ga)	Archean supercraton

Direct Textbook Quotations & Citations

B. A. van der Pluijm & S. Marshak (2004), *Earth Structure, Chapter 21, p. 542:*

*"During the Mesozoic, the supercontinent Pangea rifted along the east-west trending Tethys Ocean, separating the northern landmass comprising North America and Eurasia (***called Laurasia***) from the southern part of Pangea comprising South America, Africa, India, Australia, and Antarctica (***called Gondwana***)."*

— Van der Pluijm & Marshak, p. 542

Exam Traps & Rapid Elimination Strategy

Formula for Rapid Recall:

- **Laurentia:** North America + Greenland alone.
- **Eurasia:** Europe + Asia.
- **Laurasia = Laurentia + Eurasia:** North America + Greenland + Europe + Asia.